Soln: For all students x in the school, x has a computer or there exists another student y who has a computer and who is also a friend of x.

Simplified: Each student in the school has a computer or has a friend who has a computer.

Example 10: Translate the statement $\exists x \forall y \forall z((F(x,y) \wedge F(x,z) \wedge (y \neq z)) \rightarrow \neg F(y,z))$ to English, where $F(a,b)$ means a and b are friends and the domain for x, y and z consists of all students in your school.

Soln: There exists a student x in my school such that for any two students y and z in my school, if x is friends with both y and z and y, z are not the same person, then $\boxed{y \text{ is not a friend of } z}$.

Simplified: There exists a student in my school none of whose friends are friends with each other.

1.5.6 Translating English Sentences to Logical Expressions

Example 11: Express the statement "If a person is a female, and is a parent, then this person is someone's mother" as a logical expression involving predicates, quantifiers with a domain consisting of all people, and logical connectives.

Soln: $F(x)$: x is a female $P(x)$: x is a parent

$M(x,y)$: x is the mother of y

$$\boxed{\forall x(F(x) \wedge P(x) \Rightarrow \exists y(M(x,y)))}$$

Using the null quantification rule in part (b) of Exercise 49 in Section 1.4, we can move $\exists y$ to the left so that it appears just after $\forall x$, because y does not appear in $F(x) \wedge P(x)$. We obtain the logically equivalent expression

$$\boxed{\forall x \exists y(F(x) \wedge P(x) \Rightarrow M(x,y))}.$$